Helping Your Grade 4 Child with Number Patterns (CAPS)

A practical guide for South African parents whose children are struggling with number patterns in Grade 4 Mathematics.

Why Number Patterns Are Tricky

Number patterns require a different kind of thinking compared to straight calculation. Your child might be great at adding 37 + 28 but still struggle with patterns because patterns require:

- **Relational thinking** — seeing what connects numbers, not just computing them.
- **Working backwards** — figuring out the rule from examples, then applying it.
- **Holding multiple steps in mind** — remembering the rule while using it.

This is completely normal! These are higher-order skills that develop with practice and the right kind of support.

What CAPS Grade 4 Expects

By the end of Grade 4, your child should be able to:

Skill	Example
Count forwards/backwards in 2s, 3s, 4s, 5s, 10s, 20s, 25s, 50s, 100s	25, 50, 75, 100, 125, ...
Identify and describe the rule	"Add 5 each time"
Extend a pattern by several terms	4, 8, 12, 16, **?**, **?**
Fill in missing numbers in a sequence	7, 14, **?**, 28, 35
Complete input–output tables	If the rule is ×3, then 5 → 15
Describe patterns in own words	"It goes up by ten"

Concrete Strategies That Work

1. Start with Physical Objects

Before numbers on a page, use real things:

- **Buttons or beads:** Lay out groups of 3, then 6, then 9. Let her *see* that each group adds 3 more.
- **Building blocks:** Build towers of increasing height — 2 blocks, 4 blocks, 6 blocks. Ask "What comes next?"
- **Coins:** Use 10c, 20c, 50c pieces. "If I give you one more 10c coin each time, how much will you have?"

2. Use a Number Line

Draw a number line on paper or use a ruler:

```
0---5---10---15---20---25---30
    +5   +5   +5   +5   +5
```

The "jumps" make the rule visible and physical. Let your child draw the arches themselves.

3. Write the Differences Between Numbers

For any sequence, teach your child to write the difference between each pair:

```
 3    7    11    15    19
   +4   +4    +4    +4
```

Once the differences are written, the rule becomes obvious. This single strategy solves the majority of Grade 4 pattern questions.

4. Use the "What Changed?" Question

Train your child to always ask: **"What did they do to get from this number to the next one?"**

Practice this with simple sequences until it becomes automatic:

- 2, 4, 6, 8 → "They added 2"
- 20, 17, 14, 11 → "They subtracted 3"
- 5, 10, 15, 20 → "They added 5"

5. Connect Patterns to Times Tables

Many Grade 4 patterns are simply times tables in disguise:

Pattern	Times table
3, 6, 9, 12, 15	3× table
4, 8, 12, 16, 20	4× table
5, 10, 15, 20, 25	5× table

If your child knows their tables, point out: "This is just counting in 3s — like your 3 times table!"

6. Work with Input–Output Tables Step by Step

For tables like:

In	Out
2	6
3	9
5	?

Teach this thought process:

1. "What can I do to 2 to get 6?" (×3, or +4)
2. "Does the same thing work for 3 → 9?" (3×3=9 ✓, 3+4=7 ✗)
3. "So the rule is ×3. Now do 5 × 3 = 15."

7. Make It a Daily 5-Minute Game

Short, frequent practice beats long, occasional sessions:

- **"What comes next?"** — Say a sequence, she gives the next number.
- **"What's my rule?"** — You give a sequence, she figures out the rule.
- **"Continue my pattern"** — Start a pattern, take turns adding numbers.
- **Car game** — Use house numbers, km markers, or number plates while driving.

Common Mistakes and How to Fix Them

Mistake	Why It Happens	How to Help
Adds instead of subtracts (or vice versa)	Doesn't notice the pattern going down	Ask: "Are the numbers getting bigger or smaller?"
Gets the first missing number right but not the second	Forgets to apply the rule again	Write the rule above every gap: "+5, +5, +5"
Finds different "rules" for different pairs	Subtracts inconsistently	Practice writing all differences first, then checking they're the same
Confuses ×2 with +2	Doesn't check the rule against all pairs	Always test the rule on at least two pairs
Panics with big numbers	Intimidated by 100s and 1000s	Start with small numbers, show that the method is the same

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Daily Practice Schedule (Suggested)

Day	Activity	Time
Mon	Learn tab — read through examples together	10 min
Tue	Practice tab — easy difficulty	5-10 min
Wed	Input-output tables — easy	5-10 min
Thu	Practice tab — medium difficulty	5-10 min
Fri	60-second challenge (fun end to the week!)	5 min
Sat	Real-life patterns: money, house numbers, shopping	5 min
Sun	Rest or free choice from any tab	optional

When to Get Extra Help

If your child is still struggling after 2-3 weeks of consistent practice:

- Speak to her teacher — ask specifically which pattern types she finds hardest.
- Check that basic addition and subtraction facts are solid (patterns depend on these).
- Consider whether there's a gap in times tables knowledge (many patterns are skip-counting).
- A few sessions with a tutor who specialises in foundation phase / intermediate phase maths can make a big difference.

Using the Interactive Tool

Open `number-patterns.html` in any web browser (Chrome, Firefox, Edge, Safari). No internet connection is needed — it works completely offline.

Tabs:

1. **Learn** — Visual explanations with examples. Read through this together first.
2. **Practice** — Endless randomly-generated pattern questions at three difficulty levels.
3. **In-Out Tables** — Practice finding rules and applying them to input-output tables.
4. **Challenge** — A timed 60-second mode to build speed and confidence.

Tips for using the tool:

- Start on **Easy** and only move up when she's consistently getting 8+ correct in a row.
- Use the **Hint** button freely at first — hints teach the method.
- Celebrate streaks and challenge scores, no matter how small.
- If she gets one wrong, talk through it together before moving on.

Built with love for South African learners following the CAPS curriculum.